

YUNYONG KO

CONTACT INFORMATION	Building 310, Room 744 84, Heukseok-ro, Dongjak-gu, Seoul, 06974, Republic of Korea	Tel: + 82 - 2 - 820 - 5507 Email: yyko@cau.ac.kr Homepage: https://yy-ko.github.io
RESEARCH INTERESTS	My research interests mainly lie in data mining and machine learning on a various types of data (e.g., graph, hypergraph, text, image, and time-series), with a special focus on exploring knowledge for real-world applications.	
EDUCATION	Hanyang University , Seoul, Korea • <i>Ph.D. in Computer Science</i> – Thesis: Effective Approaches to Distributed Deep Learning: Methods, Analyses, and Evaluation – Advisor: Prof. Sang-Wook Kim Hanyang University , Seoul, Korea • <i>B.S. in Computer Science</i>	Sep. 2013 – Aug. 2021 Mar. 2009 – Aug. 2013
POSITIONS	Chung-Ang University , Seoul, Korea • <i>Assistant Professor, School of Computer Science and Engineering</i> University of Illionois at Urbana-Champaign , Urbana, IL, USA • <i>Postdoctoral Researcher, Department of Computer Science</i> – Topic: Large-Scale Hypergraph Learning for Real-World Applications – Advisor: Prof. Hanghang Tong Hanyang University , Seoul, Korea • <i>Postdoctoral Researcher, Department of Computer Science</i> – Topic: Optimization Technique for Large-Batch DNN Training – Advisor: Prof. Sang-Wook Kim The Pennsylvania State University , University Park, PA, USA • <i>Visiting Scholar, College of Information Sciences and Technology</i> – Topic: Data Parallelism Approach for Distributed Deep Learning – Advisor: Prof. Dongwon Lee	Mar. 2024 – Present May. 2022 – Feb. 2024 Sep. 2021 – April. 2022 Oct. 2019 – Feb. 2020
AWARDS & HONORS	Received the Scholarship and Teaching for Engineering Postdocs (STEP) • Grainger College of Engineering (GCOE), University of Illinois at Urbana-Champaign Selected as One of the Best-Ranked Papers • IEEE International Conference on Data Mining (IEEE ICDM) Selected as One of the Spotlight Presentations • ACM International Conference on Information and Knowledge Management (ACM CIKM) Received the Outstanding Ph.D. Dissertation Award • Research Institute of Industrial Science, Hanyang University Received the ACM SIGAPP Student Travel Award • ACM Symposium on Applied Computing (ACM SAC) Awarded the NAVER Ph.D. Fellowship • Naver Corporation Domestic Conference/Journal Awards • Best Paper Awards : KIPS Spring Confernece (2021, 2023), Journal of KIISE (2017) • Best Presentation Award : KIISE KCC (2017)	2023 2021 2021 2021 2019 2017

PUBLICATIONS

International Conference and Journal Papers (* indicates equal contributions)

- [15] CROWN: A Novel Approach to Comprehending Users' Preferences for Accurate Personalized News Recommendation
Yunyong Ko, Seongeun Ryu, and Sang-Wook Kim
WWW 2025 (*The ACM Web Conference*)
Full Paper (Acceptance Rate $\approx$ 19.8%)
Selected as an Oral Presentation of WWW 2025
- [14] Enhancing Hyperedge Prediction with Context-Aware Self-Supervised Learning
Yunyong Ko, Hanghang Tong, and Sang-Wook Kim
TKDE (*The IEEE Transactions on Knowledge and Data Engineering*)
Full Paper (SCIE Journal 2025, Q1: IF Top 1.8%)
- [13] HearHere: Mitigating Echo Chambers in News Consumption through an AI-based Web System
Youngseung Jeon, Jaehoon Kim, Sohyun Park, Yunyong Ko, Seongeun Ryu, Sang-Wook Kim, and Kyungsik Han
CSCW 2024 (*The ACM Conference on Computer-Supported Cooperative Work and Social Computing*)
Full Paper (Acceptance Rate $\approx$ 21%)
- [12] SAGE: A Storage-Based Approach for Scalable and Efficient Sparse Generalized Matrix-Matrix Multiplication {Myung-Hwan Jang*, Yunyong Ko*}, Hyuck-Moo Gwon, Ik-Hyeon Jo, Yongjun Park, and Sang-Wook Kim
CIKM 2023 (*The ACM International Conference on Information and Knowledge Management*)
Full Paper (Acceptance Rate $\approx$ 24%)
- [11] KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction
Yunyong Ko, Seongeun Ryu, Soeun Han, Youngseung Jeon, Jaehoon Kim, Sohyun Park, Kyungsik Han, Hanghang Tong and Sang-Wook Kim
WWW 2023 (*The ACM Web Conference*)
Full Paper (Acceptance Rate $\approx$ 19.2%)
- [10] RealGraph^{GPU}: A High-Performance GPU-Based Graph Engine Toward Large-Scale Real-World Network Analysis
Myung-Hwan Jang, Yunyong Ko, Dongkyu Jeong, Jeong-Min Park, and Sang-Wook Kim
CIKM 2022 (*The ACM International Conference on Information and Knowledge Management*)
Short Paper (Acceptance Rate $\approx$ 28.3%)
- [9] Not All Layers Are Equal: A Layer-Wise Adaptive Approach Toward Large-Scale DNN Training
Yunyong Ko, Dongwon Lee, and Sang-Wook Kim
WWW 2022 (*The ACM Web Conference*)
Full Paper (Acceptance Rate $\approx$ 17.7%)
- [8] D-FEND: A Diffusion-Based Fake News Detection Framework for News Articles Related to COVID-19
Soeun Han, Yunyong Ko, Yusim Kim, Heejin Park, Seongsu Oh, and Sang-Wook Kim
SAC 2022 (*The ACM Symposium on Applied Computing*)
Full Paper (Acceptance Rate $\approx$ 24%)
- [7] MASCOT: A Quantization Framework for Efficient Matrix Factorization in Recommender Systems {Yunyong Ko*, Jae-Seo Yu*}, Hong-Kyun Bae, Yongjun Park, Dongwon Lee, and Sang-Wook Kim
ICDM 2021 (*The IEEE International Conference on Data Mining*)
Full Paper (Acceptance Rate $\approx$ 9.9%)
Selected as One of the Best-ranked Papers of ICDM 2021 for Fast-track Journal Invitation
- [6] ALADDIN: Asymmetric Centralized Training for Distributed Deep Learning
Yunyong Ko, Kibong Choi, Hyunseung Je, Dongwon Lee, and Sang-Wook Kim
CIKM 2021 (*The ACM International Conference on Information and Knowledge Management*)
Full Paper (Acceptance Rate $\approx$ 21.7%)
Selected as a Spotlight Presentation of CIKM 2021

- [5] An In-Depth Analysis of Distributed Training of Deep Neural Networks
Yunyong Ko, Kibong Choi, Jiwon Seo, and Sang-Wook Kim
IPDPS 2021 (*The IEEE International Parallel and Distributed Processing Symposium*)
Full Paper (Acceptance Rate $\approx 24.5\%$)
- [4] Influence Maximization for Effective Advertisement in Social Networks: Problem, Solution, and Evaluation
Suk-Jin Hong, Yunyong Ko, Moon-Jeung Joe, and Sang-Wook Kim
SAC 2019 (*The ACM Symposium on Applied Computing*)
Full Paper (Acceptance Rate $\approx 24.2\%$)
- [3] Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach
Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
Information Sciences
Full Paper (SCIE Journal 2018, Q1: IF Top 5%)
- [2] Influence Maximization in Social Networks: A Target-Oriented Estimation
Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
Journal of Information Science
Full Paper (SCIE Journal 2018, Q2)
- [1] Accurate Path-Based Influence Maximization in Social Networks
Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
WWW 2016 (*The ACM Web Conference*)
Short Paper (Acceptance Rate $\approx 21\%$)

INVITED TALKS

Efficient and Scalable Approach for Graph Machine Learning

- Invited Talk @ Korea-Japan Database (KJDB) Workshop, Nov. 2024

Knowledge Discovery from Real-world Networks: Dilemma, Methods, and Applications

- Invited Talk @ Data Intelligence Workshop, Jul. 2024
- Invited Talk @ New Faculty Session of KCC 2024, June. 2024
- Invited Talk @ Hanyang University, Feb. 2024

KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

- Invited Talk @ Electronic & Information Research Information Center (EIRIC), Sep. 2023

Not All Layers Are Equal: A Layer-Wise Approach Towards Large-Scale DNN Training

- Poster Session @ Hyundai Vision Conference (HVC), Aug. 2023
- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

Basic Concept of Distributed Deep Learning with PyTorch Tutorials

- Invited Talk @ Medical AI Korea, Oct. 2021

Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach

- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2018
- Invited Talk @ NAVER Corp., Feb. 2018

PROFESSIONAL SERVICES

Track Co-Chair

- The ACM Symposium on Applied Computing (**SAC**) 2023 - 2025

Conference Reviewer

- The ACM Web Conference (**WWW**) 2023 - 2025
- The ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**) 2021 - 2022, 2024 - 2025
- The IEEE International Conference on Data Mining (**ICDM**) 2022, 2023
- The IEEE International Conference on Big Data (**BigData**), GTA3 Workshop 2023
- The AAAI International Conference on Artificial Intelligence (**AAAI**) 2021
- The ACM Symposium on Applied Computing (**SAC**) 2022 - 2025

Journal Reviewer

- The IEEE Transactions on Network Science and Engineering (TNSE) 2024
- The ACM Computing Surveys (CSUR) 2024
- The IEEE Transactions on Neural Networks and Learning Systems (TNNLS) 2023
- The Journal of Supercomputing 2023

PATENTS

Granted Patents

- Asymmetric Centralized Training for Distributed Deep Learning
Registration Number: **KR10-2555268** Jul. 2023
- Multi-State Diffusion Model Using Interest, Intimacy, and Share Tendency
Registration Number: **KR10-2332348** Nov. 2021
- Accurate Ad-Effect Estimation Method based on Relevance between User and Item
Registration Number: **KR10-2144122** Aug. 2020
- Hybrid Approach for Influence Maximization in Social Networks
Registration Number: **KR10-1810864** Dec. 2017

Filed Patents

- Personalized News Recommendation Technique using Category-guided Intent Disentanglement and Consistency-based News Representation
Application Number: **KR10-2023-0196006** Dec 2023
- Political Stance Prediction Method and System using Political Knowledge Graphs and Hierarchical Attention Networks
Application Number: **KR10-2023-0059346** May 2023
- Method and System for Adjusting the Learning Rate Differentiated by Layer for Large Scale Data Parallel Based Deep Learning
Application Number: **KR10-2022-0075800** June 2022

REFERENCES

- Hanghang Tong**, Associate Professor (Postdoc. Advisor) htong@illinois.edu
Department of Computer Science, University of Illinois at Urbana-Champaign (UIUC)
- Sang-Wook Kim**, Professor (Ph.D. Advisor) wook@hanyang.ac.kr
Department of Computer Science, Hanyang University
- Dongwon Lee**, Professor (Collaborator) dongwon@psu.edu
College of Information Sciences and Technology, The Pennsylvania State University (PSU)
- Kyungsik Han**, Associate Professor (Collaborator) kyungsikhan@hanyang.ac.kr
Department of Data Science, Hanyang University